Open problems:
Non-uniqueness of the moment problem for Schwartz functions

Open problems raised by the technical brief in this directory, and by its Lean formalization (see `formalization-log.tex`).

From the brief

1. **Moments see only distances to 0.** Although the topology of $\mathcal{S}(\mathbb{R})$ is generated by the seminorms $\|f\|^2_{\alpha,\beta}$, "the moments, then, can only give information about all distances to 0, and not to all other functions". This explanation of the failure is not made precise. (The seminorm equivalence itself is formalized for the brief "Equivalence of sup-norm, L^1, and L^2 seminorms on the Schwartz space".) (Brief, Remark after Theorem 1; log, "Differences from the brief": not formalized.)

2. **Locality as the obstruction.** "The key obstruction is that polynomials of X and P are local operators and can never detect phase differences between spatially separated parts." The footnote ties locality to Peetre's theorem. The general statement, that $\langle f|A|g\rangle = 0$ for every local operator A and disjointly supported f, g, so that no such A separates ψ_θ from $\psi_{\theta'}$, is not formalized. (Brief, Sec. 3 and footnote 1; log, "Differences from the brief": not formalized.)

3. **Other state spaces.** "Consequently any space that contains functions with compact support will suffer the same consequences." Not stated or proved beyond $\mathcal{S}(\mathbb{R})$, e.g. for $\mathbb{R}^n$ or other domains. (Brief, Sec. 3.)

4. **Which states are determined by their moments.** The brief asks, "As in classical probability", whether each state can be identified by its moments of polynomials in X and P, and answers only negatively. Conditions under which the moments do determine the state, the analogue of classical determinacy criteria, are not addressed. (Brief, Sec. 1.)

   Partial answer in print. Woronowicz [W1] proves a sufficient condition of the kind asked for. Let ω be a state on the algebra generated by X and P on $\mathcal{S}(\mathbb{R})$, and a, a^+ the annihilation and creation operators. If $\sum_n \sqrt{\omega(a^n(a^+)^n)}\, t^n/n! < \infty$ for some $t > 0$, then there is exactly one density operator ρ with range in $\mathcal{S}(\mathbb{R})$ such that $\omega(A) = \operatorname{Tr} A\rho$. For a pure state $|\psi\rangle$ we have $\omega(a^n(a^+)^n) = \|(a^+)^n\psi\|^2$, so the condition says that ψ is an analytic vector for a^+. The states ψ_θ of Theorem 1 cannot satisfy it, since they are not determined by their moments. The same paper contains the brief's construction: two Schwartz functions with disjoint supports give the same moments in superposition and in mixture, because "the supports of functions are never enlarged by differential operators" (Counterexample II, credited to J. Plebański). The sequel [W2] shows that uniqueness is a property of the algebra: on the algebra of all operators that map $\mathcal{S}(\mathbb{R})$ into itself, together with their adjoints, every positive functional is $\operatorname{Tr} A\rho$ with a unique ρ. Its Theorem 6 characterizes uniqueness on the polynomial algebra

(statement read, proof not checked). Open: a necessary and sufficient condition in terms of ψ alone, and the case of $\mathbb{R}^n$.

[W1] S. L. Woronowicz, The quantum problem of moments I, Rep. Math. Phys. **1** (1970) 135–145.

[W2] S. L. Woronowicz, The quantum problem of moments II, Rep. Math. Phys. **1** (1971) 175–183.

From the formalization

1. None found.